

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 11 • Trigonometric Functions

With Answers

Questions with full worked solutions

- 46 pages
- Past papers up to June 2026

Trigonometric Functions

Your classified route

16

questions

97

marks

4

families

11

variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Trigonometric Functions

Families and variants

F01 • Reading and identifying trigonometric graphs

V01 Key points, range and period

V02 Infer amplitude, frequency or equation from features

V03 Key points under vertical, horizontal and reflection transformations

F02 • Symmetry, reference angles and symbolic solutions

V01 Reference-angle and sign identities

V02 Solve a trigonometric equation in terms of alpha

V03 Link sine and cosine solutions by phase

F03 • Counting roots and graphical intersections

V01 Count solutions of a level equation over long intervals

V02 Count intersections with a line or non-trigonometric curve

V03 Parameter values giving an exact root count

F04 • Constructing trigonometric sketches

V01 Sketch transformed sine and cosine

V02 Sketch tangent with vertical asymptotes

Reading and identifying trigonometric graphs

WHAT THIS FAMILY ASKS

Use amplitude, period, sign, phase and key points to read a graph or infer its equation and parameters.

VARIANT ROUTE

V02 Infer amplitude, frequency or equation from features

V03 Key points under vertical, horizontal and reflection transformations

9 Complete Classified questions

54 total marks

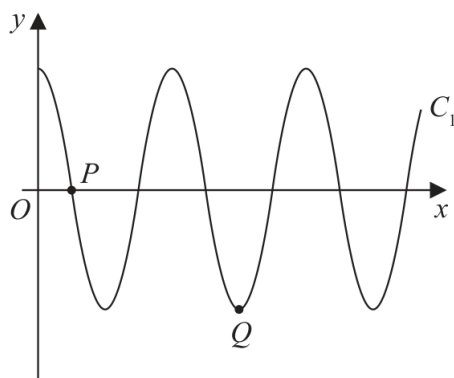


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 3 \cos\left(\frac{x}{n}\right)^\circ \quad x \geq 0$$

where n is a constant.

The curve C_1 cuts the positive x -axis for the first time at point $P(270, 0)$, as shown in Figure 4.

(a) (i) State the value of n

(ii) State the period of C_1

(2)

The point Q , shown in Figure 4, is a minimum point of C_1

(b) State the coordinates of Q .

(2)

The curve C_2 has equation $y = 2 \sin x^\circ + k$, where k is a constant.

The point $R\left(a, \frac{12}{5}\right)$ and the point $S\left(-a, -\frac{3}{5}\right)$, both lie on C_2

Given that a is a constant less than 90

(c) find the value of k .

(2)

(Total for Question 10 is 6 marks)

Question 1

WMA11/01 October 2023

T11 Trigonometric Functions

F01 Reading and identifying trigonometric graphs

V02 Infer amplitude, frequency or equation from features

1

Use the first positive x-interceptFor $y = 3 \cos\left(\frac{x}{n}\right)^\circ$, the first positive zero occurs when $\frac{x}{n} = 90^\circ$.

2

Find nSince $P = (270, 0)$, $\frac{270}{n} = 90$, so $n = 3$.

3

Find the periodThe period is $360n = 360(3) = 1080^\circ$.

4

Read the shown minimum pointA minimum occurs when $\frac{x}{3} = 180^\circ + 360^\circ k$. The point Q shown is the next displayed minimum after one full period, so $x = 1620$ and $y = -3$.

5

Use the two points on C2For $y = 2 \sin x^\circ + k$, the points give $2 \sin a + k = \frac{12}{5}$ and $-2 \sin a + k = -\frac{3}{5}$.

6

Add the equationsAdding gives $2k = \frac{12}{5} - \frac{3}{5} = \frac{9}{5}$, so $k = \frac{9}{10}$.

FINAL ANSWER

(a)(i) $n = 3$. (a)(ii) 1080° . (b) $Q = (1620, -3)$. (c) $k = \frac{9}{10}$.

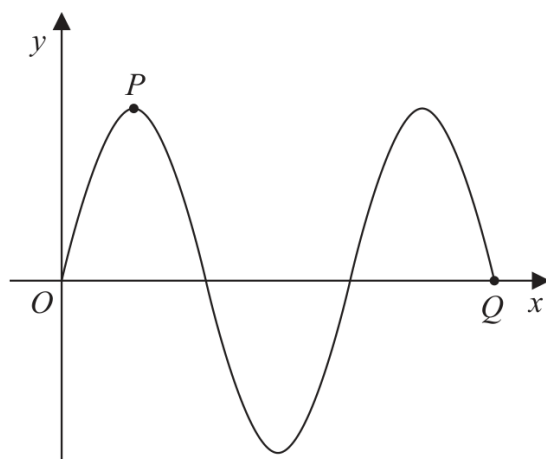


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 3 \sin x$, where x is measured in degrees.

The point P and the point Q lie on C_1 and are shown in Figure 3.

(a) State

- (i) the coordinates of P ,
- (ii) the coordinates of Q .

(3)

A different curve C_2 has equation $y = 3 \sin x + k$, where k is a constant.

The curve C_2 has a maximum y value of 10

The point R is the minimum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)

(Total 5 marks)

Question 2

WMA11/01 January 2020

T11 Trigonometric Functions

F01 Reading and identifying
trigonometric graphsV03 Key points under vertical,
horizontal and reflection trans-
formations

1

Read point P

The first maximum of $y = 3 \sin x$ occurs at $x = 90^\circ$, with $y = 3$. So $P = (90^\circ, 3)$.

2

Read Q from the next shown zero

After the second maximum, the curve returns to the x-axis at $x = 540^\circ$. So $Q = (540^\circ, 0)$.

3

Find the vertical shift

For C_2 , $y = 3 \sin x + k$. Its maximum is $3 + k = 10$, so $k = 7$.

4

Find the minimum point R

The minimum value is $-3 + 7 = 4$. The smallest positive x-coordinate for a sine minimum is 270° , so $R = (270^\circ, 4)$.

FINAL ANSWER

(a)(i) $P = (90^\circ, 3)$. (a)(ii) $Q = (540^\circ, 0)$. (b) $R = (270^\circ, 4)$.

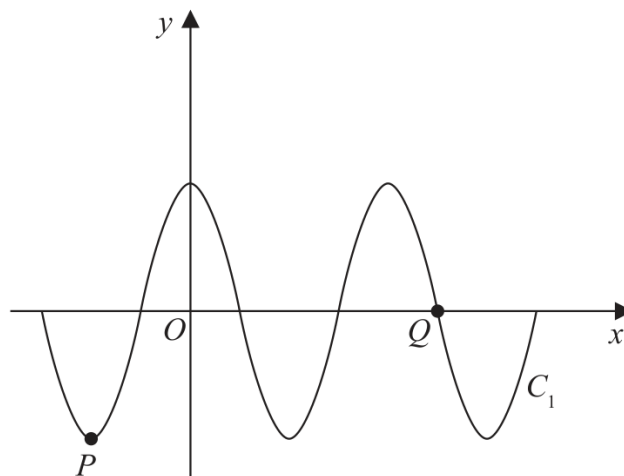


Figure 1

Figure 1 shows a sketch of part of the curve C_1 with equation $y = 4 \cos x^\circ$

The point P and the point Q lie on C_1 and are shown in Figure 1.

(a) State

(i) the coordinates of P ,

(ii) the coordinates of Q .

(3)

The curve C_2 has equation $y = 4 \cos x^\circ + k$, where k is a constant.

Curve C_2 has a minimum y value of -1

The point R is the maximum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)

(Total 5 marks)

Question 3

WMA11/01 January 2021

T11 Trigonometric Functions

F01 Reading and identifying trigonometric graphs

V03 Key points under vertical, horizontal and reflection transformations

1

Read cosine key points

For $y = 4 \cos x^\circ$, the maximum value is 4 and the minimum value is -4 . The minimum immediately to the left of the y -axis is at $x = -180^\circ$, so $P = (-180, -4)$.

2

Find Q from the cosine zero

The x -intercepts of $4 \cos x^\circ$ occur at $x = 90^\circ, 270^\circ, 450^\circ, \dots$. The point shown to the right of the next maximum is $Q = (450, 0)$.

3

Use the shifted curve C2

For C_2 , $y = 4 \cos x^\circ + k$. Its minimum is $-4 + k$. Since the minimum value is -1 , $-4 + k = -1$, so $k = 3$.

4

Find R

The maximum value of C_2 is $4 + 3 = 7$. The maximum point with the smallest positive x -coordinate occurs at $x = 360^\circ$, so $R = (360, 7)$.

FINAL ANSWER

(a)(i) $P = (-180, -4)$. (a)(ii) $Q = (450, 0)$. (b) $R = (360, 7)$.

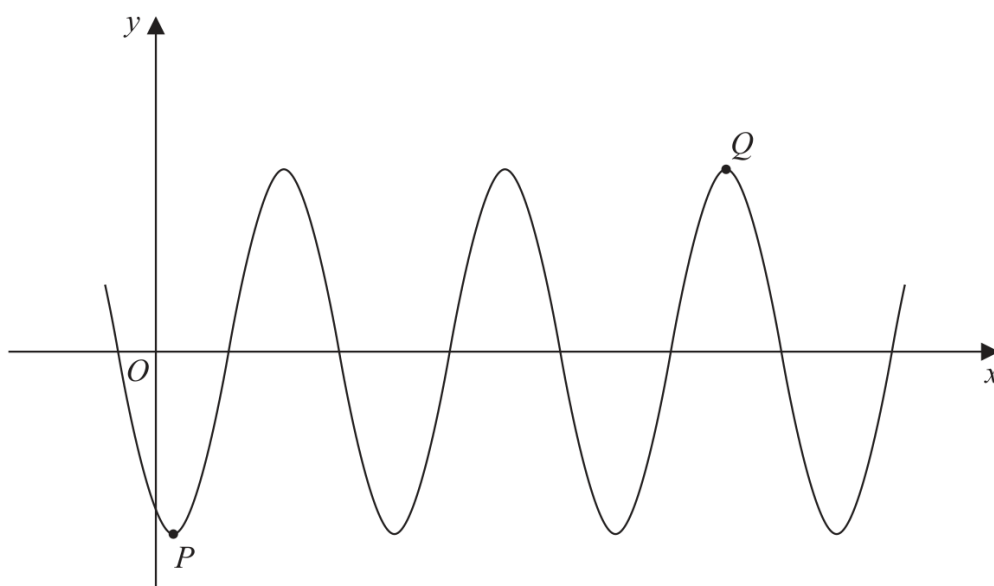


Figure 4

Figure 4 shows part of the curve with equation

$$y = A \cos(x - 30)^\circ$$

where A is a constant.

The point P is a minimum point on the curve and has coordinates $(30, -3)$ as shown in Figure 4.

(a) Write down the value of A .

(1)

The point Q is shown in Figure 4 and is a maximum point.

(b) Find the coordinates of Q .

(3)

(Total 4 marks)

Question 4

WMA11/01 January 2022

T11 Trigonometric Functions

F01 Reading and identifying
trigonometric graphsV03 Key points under vertical,
horizontal and reflection trans-
formations

1

Use the minimum point

At $P(30, -3)$, $x - 30 = 0$, so $\cos 0^\circ = 1$. Therefore $A = -3$.

2

Find maximum points

Since $A = -3$, maximum points occur when $\cos(x - 30)^\circ = -1$.

3

Use the shown maximum Q

$x - 30 = 180 + 360n$, so $x = 210 + 360n$. The maximum shown at Q is two full periods after the first maximum, so $x = 210 + 720 = 930$.

4

Write Q

The maximum y-value is 3, hence $Q = (930, 3)$.

FINAL ANSWER

(a) $A = -3$. (b) $Q = (930, 3)$.

9. (i)

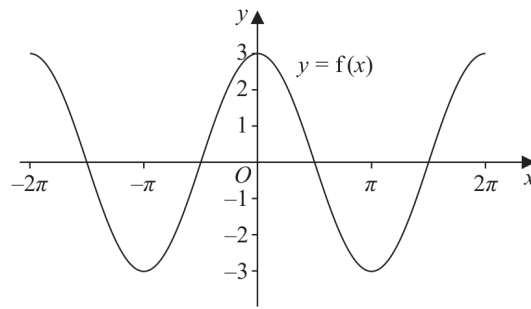


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = f\left(x + \frac{\pi}{4}\right)$

Show clearly the coordinates of all the points where the curve intersects the coordinate axes.

(3)

(ii)

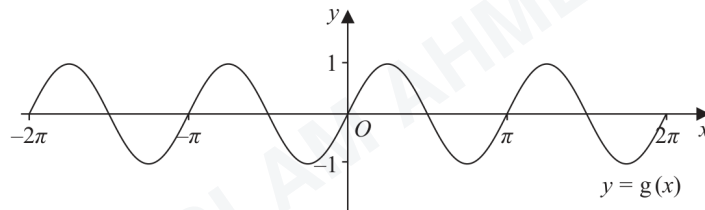


Figure 4

Figure 4 shows part of the graph of the trigonometric function with equation $y = g(x)$

(a) Write down an expression for $g(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = g(x) - 2$

Show clearly the coordinates of the y intercept.

(2)

(Total for Question 9 is 9 marks)

Question 5

WMA11/01 May/June 2023

T11 Trigonometric Functions

F01 Reading and identifying trigonometric graphs

V03 Key points under vertical, horizontal and reflection transformations

1

Identify $f(x)$

The first graph has amplitude 3, period 2π , and maximum at $x = 0$, so $f(x) = 3 \cos x$.

2

Transform f

$y = f\left(x + \frac{\pi}{4}\right) = 3 \cos\left(x + \frac{\pi}{4}\right)$, a shift left by $\frac{\pi}{4}$.

3

Find the x-intercepts

$\cos\left(x + \frac{\pi}{4}\right) = 0$ gives $x = \frac{\pi}{4} + n\pi$. In $-2\pi < x < 2\pi$, these are $-\frac{7\pi}{4}$, $-\frac{3\pi}{4}$, $\frac{\pi}{4}$, and $\frac{5\pi}{4}$.

4

Find the y-intercept

At $x = 0$, $y = 3 \cos \frac{\pi}{4} = \frac{3\sqrt{2}}{2}$.

5

Identify $g(x)$

The second graph has amplitude 1, period π , and passes through the origin with positive gradient, so $g(x) = \sin 2x$.

6

Transform g

$y = g(x) - 2 = \sin 2x - 2$, so the graph is shifted down by 2. Its y-intercept is $(0, -2)$.

FINAL ANSWER

(i)(a) $f(x) = 3 \cos x$. (i)(b) **x-intercepts** $-\frac{7\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}$ and **y-intercept** $\frac{3\sqrt{2}}{2}$. (ii)(a) $g(x) = \sin 2x$. (ii)(b) **y-intercept** $(0, -2)$.

DR ESLAM AHMED

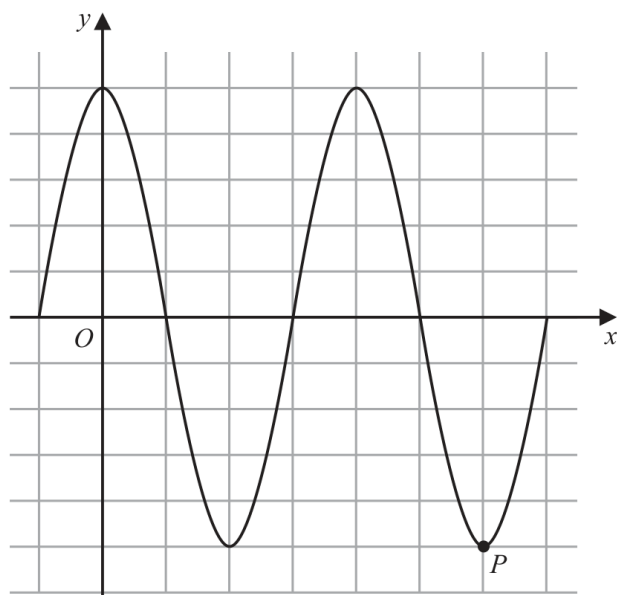


Figure 2

Figure 2 shows a plot of part of the curve C_1 with equation

$$y = 5 \cos x$$

with x being measured in degrees.

The point P , shown in Figure 2, is a minimum point on C_1

(a) State the coordinates of P

(2)

The point Q lies on a different curve C_2

Given that point Q

- is a maximum point on the curve
- is the maximum point with the **smallest** x coordinate, $x > 0$

(b) find the coordinates of Q when

(i) C_2 has equation $y = 5 \cos x - 2$

(ii) C_2 has equation $y = -5 \cos x$

(4)

(Total for Question 6 is 6 marks)

Question 6

WMA11/01 January 2024

T11 Trigonometric Functions

F01 Reading and identifying trigonometric graphs

V03 Key points under vertical, horizontal and reflection transformations

1

Use the cosine graphFor $y = 5 \cos x$, the minimum value is -5 and it occurs at $x = 180^\circ + 360^\circ n$.

2

Read P from the displayed minimumThe plotted point P is the displayed minimum at $x = 540^\circ$, so $P = (540^\circ, -5)$.

3

Translate the graph downFor $y = 5 \cos x - 2$, the maximum value is $5 - 2 = 3$. The smallest positive x-coordinate for a maximum shown is $x = 360^\circ$, so $Q = (360^\circ, 3)$.

4

Reflect in the x-axisFor $y = -5 \cos x$, a maximum occurs where $\cos x = -1$, first at $x = 180^\circ$. The maximum value is 5 , so $Q = (180^\circ, 5)$.**FINAL ANSWER**(a) $P = (540^\circ, -5)$. (b)(i) $Q = (360^\circ, 3)$. (b)(ii) $Q = (180^\circ, 5)$.

11.

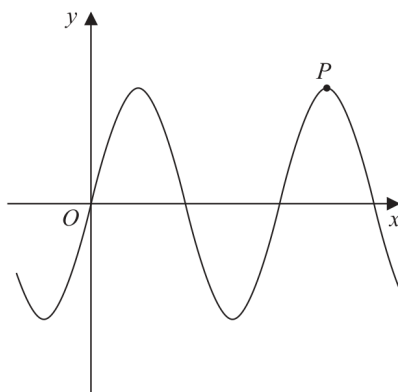


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 12 \sin x$$

where x is measured in radians.

The point P shown in Figure 4 is a maximum point on C_1

(a) Find the coordinates of P .

(2)

The curve C_2 has equation

$$y = 12 \sin x + k$$

where k is a constant.

Given that the **maximum** value of y on C_2 is 3

(b) find the coordinates of the **minimum** point on C_2 which has the **smallest** positive x coordinate.

(2)

The curve C_3 has equation

$$y = 12 \sin(x + B)$$

where B is a positive constant.

Given that $\left(\frac{\pi}{4}, A\right)$, where A is a constant, is the **minimum** point on C_3 which has the **smallest** positive x coordinate,

(c) find

(i) the value of A ,

(ii) the smallest possible value of B .

(2)

(Total for Question 11 is 6 marks)

Question 7

WMA11/01 May/June 2024

T11 Trigonometric Functions

F01 Reading and identifying trigonometric graphs

V03 Key points under vertical, horizontal and reflection transformations

1

Use maxima of sineFor $y = 12 \sin x$, maxima occur when $x = \frac{\pi}{2} + 2\pi n$, with value 12.

2

Read point PThe maximum point P shown is the second positive maximum, so $P = (\frac{5\pi}{2}, 12)$.

3

Find k for C2For $y = 12 \sin x + k$, the maximum value is $12 + k$. Given the maximum is 3, $k = -9$.

4

Find the smallest positive minimum on C2The minimum value is $-12 - 9 = -21$, and it first occurs at $x = \frac{3\pi}{2}$. So the point is $(\frac{3\pi}{2}, -21)$.

5

Find A for C3For $y = 12 \sin(x + B)$, any minimum has value -12 , so $A = -12$.

6

Find the smallest positive B At the minimum $x = \frac{\pi}{4}$, so $\frac{\pi}{4} + B = \frac{3\pi}{2}$. Hence $B = \frac{5\pi}{4}$.

FINAL ANSWER

$$(a) P = \left(\frac{5\pi}{2}, 12\right). \quad (b) \left(\frac{3\pi}{2}, -21\right). \quad (c)(i) A = -12. \quad (c)(ii) B = \frac{5\pi}{4}.$$

DR ESLAM AHMED

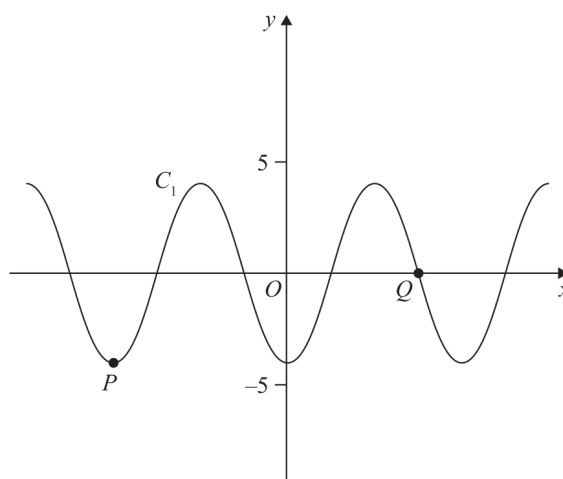


Figure 3

Figure 3 shows a plot of part of the curve C_1 with equation

$$y = -4 \cos x$$

where x is measured in radians.

Points P and Q lie on the curve and are shown in Figure 3.

(a) State

- (i) the coordinates of P
- (ii) the coordinates of Q

(3)

The curve C_2 has equation $y = -4 \cos x + k$ where x is measured in radians and k is a constant.

Given that C_2 has a maximum y value of 11

- (b) (i) state the value of k
- (ii) state the coordinates of the minimum point on C_2 with the smallest positive x coordinate.

(3)

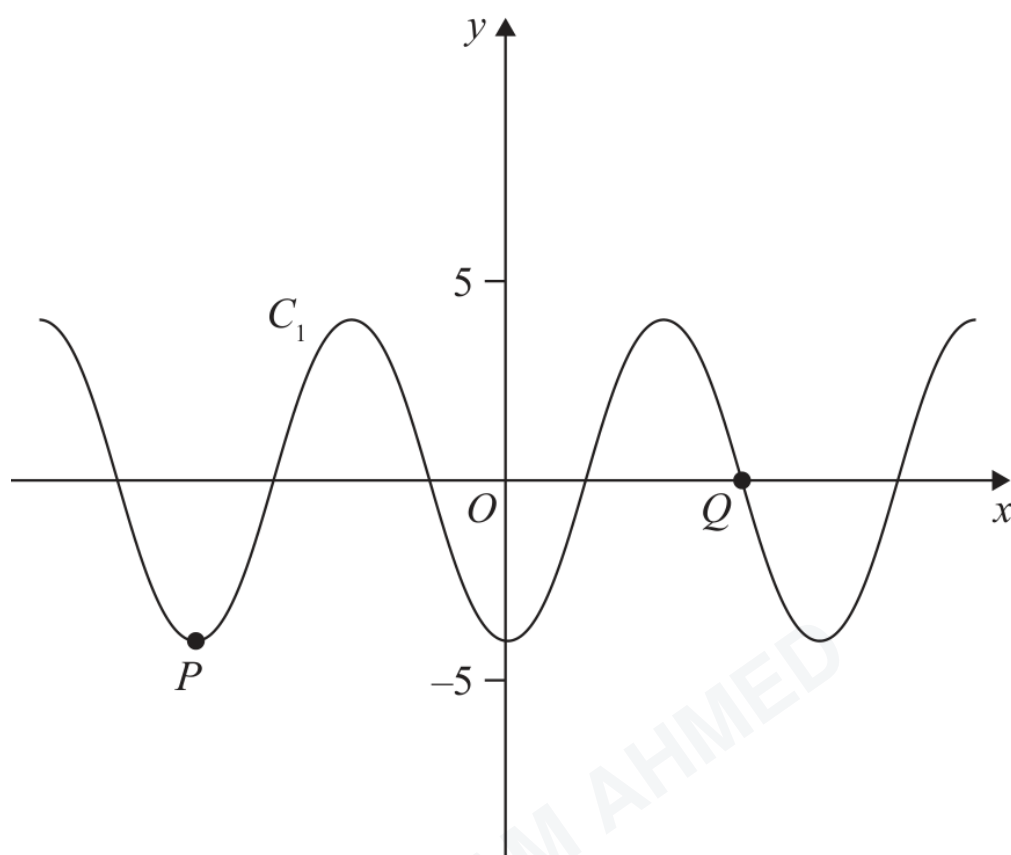
On the opposite page there is a copy of Figure 3 labelled Diagram 1.

(c) Using Diagram 1, state the number of solutions of the equation

$$-4 \cos x = 5 - \frac{10}{\pi} x$$

giving a reason for your answer.

(2)

**Diagram 1**

(Total for Question 7 is 8 marks)

Question 8

WMA11/01 October 2024

T11 Trigonometric Functions

F01 Reading and identifying trigonometric graphs

V03 Key points under vertical, horizontal and reflection transformations

1

Use y equals negative $4 \cos x$ For $y = -4 \cos x$, minimum points occur at $x = 2\pi n$ with value -4 .

2

Read PThe displayed minimum to the left of the y -axis is $P = (-2\pi, -4)$.

3

Read QThe point Q is the positive x -intercept after the maximum at $x = \pi$, so $Q = (\frac{3\pi}{2}, 0)$.

4

Find k for C2For $y = -4 \cos x + k$, the maximum value is $4 + k$. Given the maximum is 11, $k = 7$.

5

Find the smallest positive minimum on C2The minimum value is $-4 + 7 = 3$. The first positive minimum is at $x = 2\pi$, so the point is $(2\pi, 3)$.

6

Interpret the equation graphically $-4 \cos x = 5 - \frac{10}{\pi}x$ is the intersection of the cosine curve with the straight line $y = 5 - \frac{10}{\pi}x$.

7

Count intersections

The line passes through $(0,5)$ and meets the displayed curve once, at the x-axis point, so there is **1 solution**.

FINAL ANSWER

(a)(i) $P = (-2\pi, -4)$. (a)(ii) $Q = (\frac{3\pi}{2}, 0)$. (b)(i) $k = 7$.
(b)(ii) $(2\pi, 3)$. (c) **1 solution**.

DR ESLAM AHMED

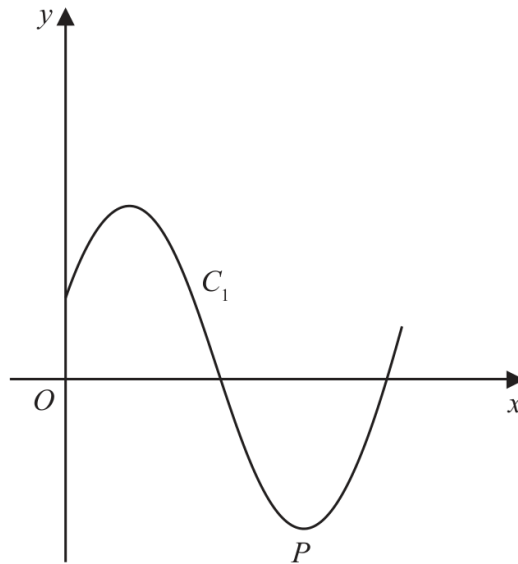


Figure 4

The curve C_1 has equation $y = \sin\left(x + \frac{\pi}{8}\right)$

Figure 4 shows a sketch of C_1 in the interval $0 \leq x < 2\pi$

The point P is a minimum turning point on C_1 and is shown in Figure 4.

(a) State the x coordinate of P .

(1)

(b) State the number of roots of the equation

$$\sin\left(x + \frac{\pi}{8}\right) = \sin\left(\frac{\pi}{8}\right)$$

in the interval $0 \leq x < 101\pi$, giving a reason for your answer.

(2)

The curve C_2 has equation $y = -5\sin\left(x + \frac{\pi}{8}\right)$

The point Q is the minimum turning point on C_2 in the interval $0 \leq x < 2\pi$

(c) State the coordinates of Q .

(2)

(Total for Question 9 is 5 marks)

Question 9

WMA11/01 May/June 2026

T11 Trigonometric Functions

F01 Reading and identifying trigonometric graphs

V03 Key points under vertical, horizontal and reflection transformations

1

(a) Locate the minimum of C1A sine curve reaches its minimum when its angle is $\frac{3\pi}{2}$. Therefore

$$x + \frac{\pi}{8} = \frac{3\pi}{2},$$

so

$$x = \frac{12\pi}{8} - \frac{\pi}{8} = \frac{11\pi}{8}.$$

2

(b) Write both families of rootsFor $\sin u = \sin\left(\frac{\pi}{8}\right)$,

$$u = \frac{\pi}{8} + 2n\pi \quad \text{or} \quad u = \frac{7\pi}{8} + 2n\pi.$$

With $u = x + \frac{\pi}{8}$, this gives

$$x = 2n\pi \quad \text{or} \quad x = \frac{3\pi}{4} + 2n\pi.$$

3

(b) Count the roots in the intervalFor $0 \leq x < 101\pi$, each family has $n = 0, 1, \dots, 50$, giving 51 roots from each family. Hence the total number of roots is

$$51 + 51 = 102.$$

4

(c) Locate the minimum of C2

Because $y = -5 \sin\left(x + \frac{\pi}{8}\right)$, the minimum value -5 occurs when the sine value is 1. Thus

$$x + \frac{\pi}{8} = \frac{\pi}{2},$$

so $x = \frac{3\pi}{8}$. Therefore

$$Q = \left(\frac{3\pi}{8}, -5\right).$$

FINAL ANSWER

$$(a) x = \frac{11\pi}{8}. \quad (b) 102 \text{ roots.} \quad (c) Q = \left(\frac{3\pi}{8}, -5\right).$$

DR ESAM AHMED

Symmetry, reference angles and symbolic solutions

WHAT THIS FAMILY ASKS

Use trigonometric symmetry and phase links to express exact values or solutions in terms of a given angle α .

VARIANT ROUTE

V02 Solve a trigonometric equation in terms of α

V03 Link sine and cosine solutions by phase

2 Complete Classified questions

12 total marks

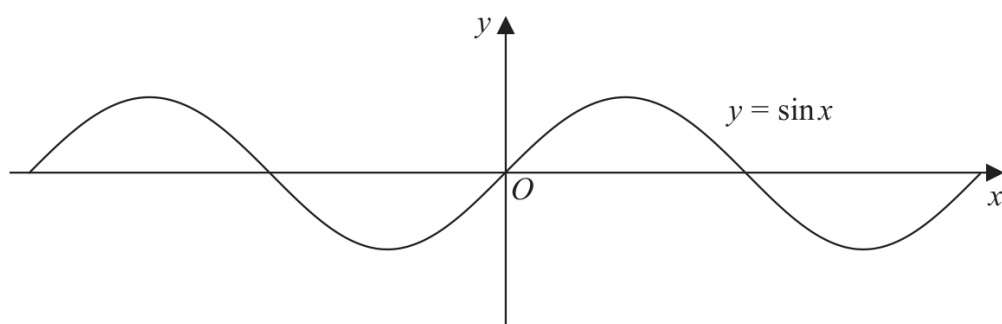


Figure 4

Figure 4 shows part of the graph of the curve with equation $y = \sin x$

Given that $\sin \alpha = p$, where $0 < \alpha < 90^\circ$

(a) state, in terms of p , the value of

(i) $2 \sin(180^\circ - \alpha)$

(ii) $\sin(\alpha - 180^\circ)$

(iii) $3 + \sin(180^\circ + \alpha)$

(3)

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

(b) sketch the graph of $y = \sin 2x$

(2)

(c) Hence find, in terms of α , the x coordinates of any points in the interval $0 < x < 180^\circ$ where

$$\sin 2x = p$$

(3)

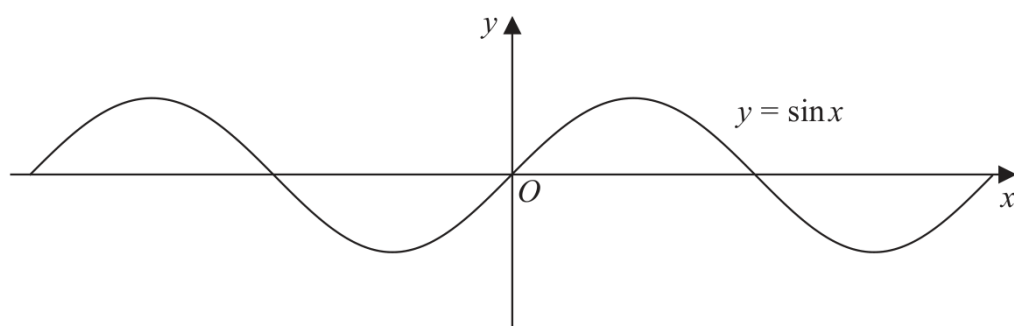


Diagram 1

(Total 8 marks)

DR ESLAM AHMED

Question 10

WMA11/01 May/June 2022

T11 Trigonometric Functions

F02 Symmetry, reference angles and symbolic solutions

V02 Solve a trigonometric equation in terms of alpha

1

Use sine symmetry $\sin(180^\circ - \alpha) = \sin \alpha = p$, so $2 \sin(180^\circ - \alpha) = 2p$.

2

Use odd symmetry $\sin(\alpha - 180^\circ) = -\sin(180^\circ - \alpha) = -p$.

3

Use the third quadrant identity $\sin(180^\circ + \alpha) = -\sin \alpha = -p$, so $3 + \sin(180^\circ + \alpha) = 3 - p$.

4

Sketch $y = \sin 2x$ The graph has amplitude 1 and period 180° . It crosses the x-axis at $0^\circ, 90^\circ, 180^\circ$, has a maximum at 45° , and a minimum at 135° .

5

Solve $\sin 2x = p$ Since $\sin \alpha = p$ and $0 < 2x < 360^\circ$, the possible values are $2x = \alpha$ or $2x = 180^\circ - \alpha$.

6

Find x Therefore $x = \frac{\alpha}{2}$ or $x = 90^\circ - \frac{\alpha}{2}$.**FINAL ANSWER**(a) (i) $2p$, (ii) $-p$, (iii) $3 - p$. (c) $x = \frac{\alpha}{2}, 90^\circ - \frac{\alpha}{2}$.

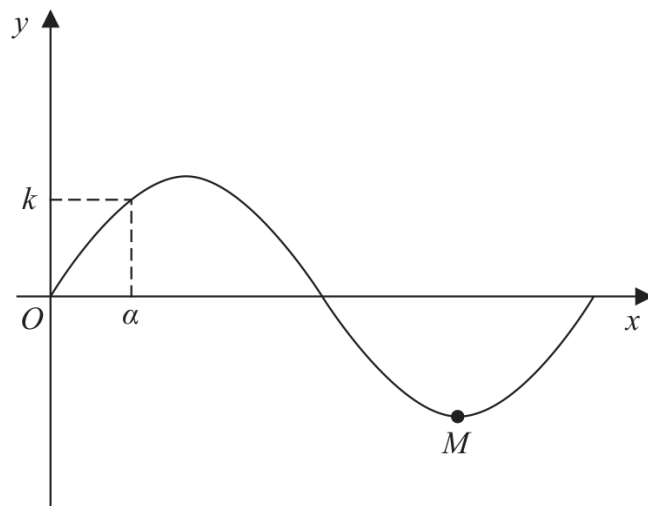


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C .

(a) State the period of C . (1)

(b) State the coordinates of M . (1)

The smallest positive solution of the equation $\sin\left(\frac{x}{12}\right) = k$, where k is a constant, is α .

Find, in terms of α ,

(c) (i) the negative solution of the equation $\sin\left(\frac{x}{12}\right) = k$ that is closest to zero,
(ii) the smallest positive solution of the equation $\cos\left(\frac{x}{12}\right) = k$. (2)

(Total 4 marks)

Question 11

WMA11/01 October 2019

T11 Trigonometric Functions

F02 Symmetry, reference angles and symbolic solutions

V03 Link sine and cosine solutions by phase

1

Use the scale factor inside the sine

For $y = \sin\left(\frac{x}{12}\right)$, one full sine cycle needs $\frac{x}{12}$ to increase by 2π , so the period is 24π .

2

Locate the minimum point

The sine function is -1 when its angle is $\frac{3\pi}{2}$. Therefore $\frac{x}{12} = \frac{3\pi}{2}$, so $x = 18\pi$ and $M = (18\pi, -1)$.

3

Use sine symmetry

If the smallest positive solution of $\sin\left(\frac{x}{12}\right) = k$ is α , then the closest negative solution is one half-period before the opposite sine position: $-12\pi - \alpha$.

4

Convert to cosine

Since $\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$, the corresponding smallest positive cosine solution is $6\pi - \alpha$.

FINAL ANSWER

(a) 24π . (b) $(18\pi, -1)$. (c)(i) $-12\pi - \alpha$. (ii) $6\pi - \alpha$.

Counting roots and graphical intersections

WHAT THIS FAMILY ASKS

Count solutions from periodic graphs, endpoint rules and intersections with horizontal lines or non-trigonometric curves.

VARIANT ROUTE

V01 Count solutions of a level equation over long intervals

V02 Count intersections with a line or non-trigonometric curve

V03 Parameter values giving an exact root count

5 Complete Classified questions

31 total marks

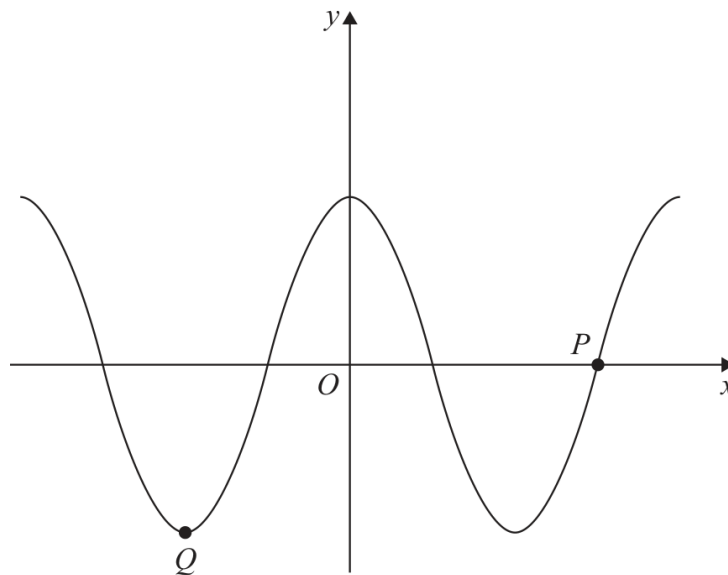


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = 4 \cos x$$

where x is measured in degrees.

The points P and Q lie on the curve and are shown in Figure 1.

- (a) State the coordinates of P . (1)
- (b) State the coordinates of Q . (2)
- (c) State the **number** of solutions of the equation
 - (i) $4 \cos x = 3$ in the interval $0 < x \leq 18\,000^\circ$
 - (ii) $5 + 4 \cos x = 1$ in the interval $-720^\circ < x \leq 720^\circ$
 - (iii) $4 \cos x - 3 = 1$ in the interval $-1080^\circ \leq x \leq 1080^\circ$ (3)

(Total for Question 4 is 6 marks)

Question 12

WMA11/01 May/June 2025

T11 Trigonometric Functions

F03 Counting roots and graphical intersections

V01 Count solutions of a level equation over long intervals

1

Point P

For $y = 4 \cos x$, the x-intercepts occur at $x = 90^\circ + 180^\circ n$. The labelled point P is the right-hand intercept after 180° , so $P = (270^\circ, 0)$.

2

Point Q

The minimum value is -4 , reached at $x = 180^\circ + 360^\circ n$. The labelled point on the left is $Q = (-180^\circ, -4)$.

3

Count part c(i)

$4 \cos x = 3$ has two solutions every 360° . The interval $0 < x \leq 18000^\circ$ contains 50 full periods, so there are 100 solutions.

4

Count part c(ii)

$5 + 4 \cos x = 1$ gives $\cos x = -1$. This occurs once every 360° . In $-720^\circ < x \leq 720^\circ$, the solutions are $-540^\circ, -180^\circ, 180^\circ, 540^\circ$, so there are 4.

5

Count part c(iii)

$4 \cos x - 3 = 1$ gives $\cos x = 1$. In $-1080^\circ \leq x \leq 1080^\circ$, the multiples of 360° are $-1080, -720, -360, 0, 360, 720, 1080$, so there are 7.

FINAL ANSWER

(a) $P = (270^\circ, 0)$. (b) $Q = (-180^\circ, -4)$. (c) (i) 100, (ii) 4, (iii) 7.

7.

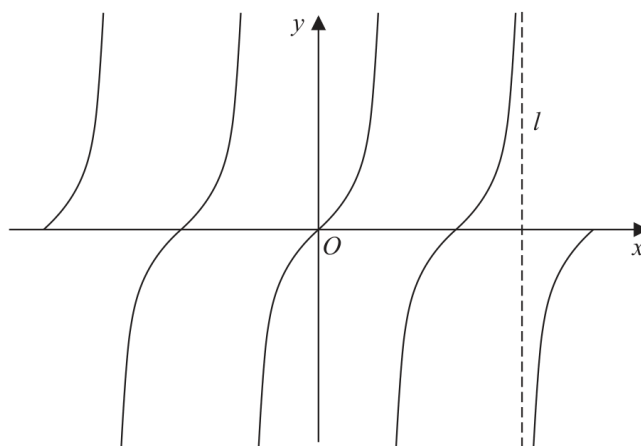


Figure 4

Figure 4 shows a sketch of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

The line l , shown in Figure 4, is an asymptote to $y = \tan x$

(a) State an equation for l .

(1)

A copy of Figure 4, labelled Diagram 1, is shown on the next page.

(b) (i) On Diagram 1, sketch the curve with equation

$$y = \frac{1}{x} + 1 \quad -2\pi \leq x \leq 2\pi$$

stating the equation of the horizontal asymptote of this curve.

(ii) Hence, **giving a reason**, state the number of solutions of the equation

$$\tan x = \frac{1}{x} + 1$$

in the region $-2\pi \leq x \leq 2\pi$

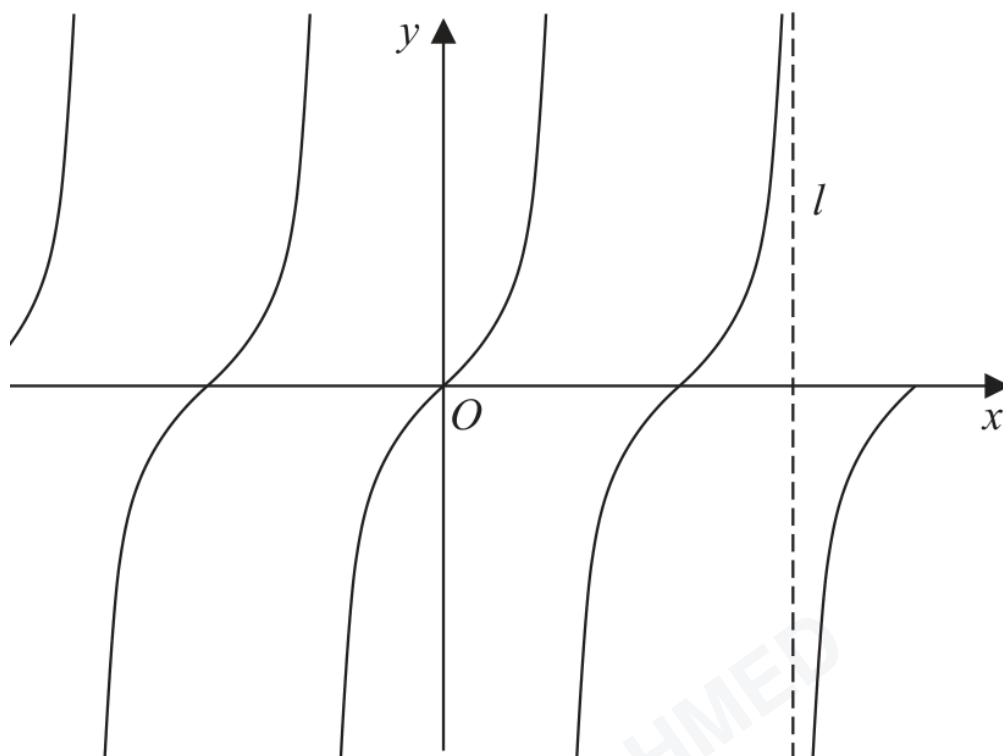
(4)

(c) State the number of solutions of the equation $\tan x = \frac{1}{x} + 1$ in the region

(i) $0 \leq x \leq 40\pi$

(ii) $-10\pi \leq x \leq \frac{5}{2}\pi$

(2)

**Diagram 1**

(Total 7 marks)

Question 13

WMA11/01 May/June 2021

T11 Trigonometric Functions

F03 Counting roots and graphical intersections

V02 Count intersections with a line or non-trigonometric curve

1

State the shown tangent asymptote

The shown dashed asymptote is the right-hand positive asymptote before 2π , so l has equation $x = \frac{3\pi}{2}$.

2

Describe the reciprocal graph

For $y = \frac{1}{x} + 1$, the vertical asymptote is $x = 0$ and the horizontal asymptote is $y = 1$. The right branch lies above $y = 1$ and the left branch lies below $y = 1$.

3

Count intersections on -2π to 2π

The number of solutions of $\tan x = \frac{1}{x} + 1$ is the number of intersections of the two graphs. From the sketch on $-2\pi \leq x \leq 2\pi$, there are 5 intersections.

4

Count first interval solutions

From the repeated tangent branches and the reciprocal curve, there is one intersection in each positive interval between consecutive tangent asymptotes. From 0 to 40π , this gives 40 solutions.

5

Count second interval solutions

Extending the same intersection pattern across the stated interval gives 14 intersections, so there are 14 solutions.

FINAL ANSWER

(a) $x = \frac{3\pi}{2}$. (b)(i) $y = 1$ is the horizontal asymptote. (b)(ii) 5.
(c)(i) 40. (c)(ii) 14.

9.

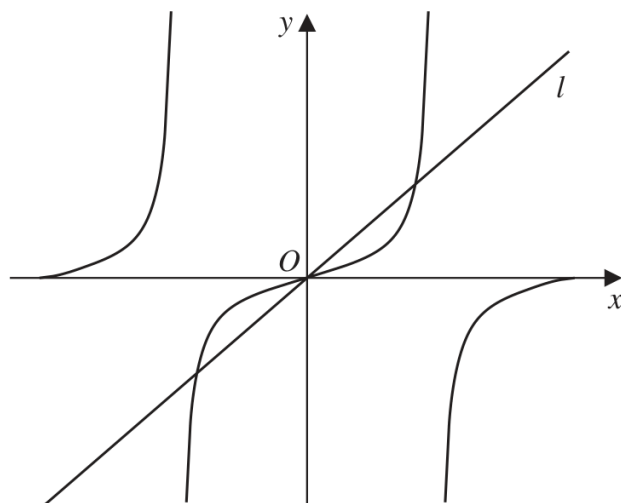


Figure 3

Figure 3 shows a sketch of

- the curve with equation $y = \tan x$
- the straight line l with equation $y = \pi x$

in the interval $-\pi < x < \pi$

(a) State the period of $\tan x$

(1)

(b) Write down the number of roots of the equation

(i) $\tan x = (\pi + 2)x$ in the interval $-\pi < x < \pi$

(1)

(ii) $\tan x = \pi x$ in the interval $-2\pi < x < 2\pi$

(1)

(iii) $\tan x = \pi x$ in the interval $-100\pi < x < 100\pi$

(1)

(Total for Question 9 is 4 marks)

Question 14

WMA11/01 January 2023

T11 Trigonometric Functions

F03 Counting roots and graphical intersections

V02 Count intersections with a line or non-trigonometric curve

1

State the periodThe period of $\tan x$ is π .

2

Count roots for the steeper lineIn $-\pi < x < \pi$, the line $y = (\pi + 2)x$ meets $y = \tan x$ once at the origin and once on each side of the origin, giving 3 roots.

3

Extend to $-2\pi < x < 2\pi$ For $y = \pi x$, the interval $-2\pi < x < 2\pi$ has the 3 central roots plus one further root on each neighbouring branch, giving 5 roots.

4

Extend to $-100\pi < x < 100\pi$ There are 99 positive outer roots, 99 negative outer roots, and 3 central roots, so the total is $99 + 99 + 3 = 201$.**FINAL ANSWER**(a) π . (b)(i) 3, (ii) 5, (iii) 201.

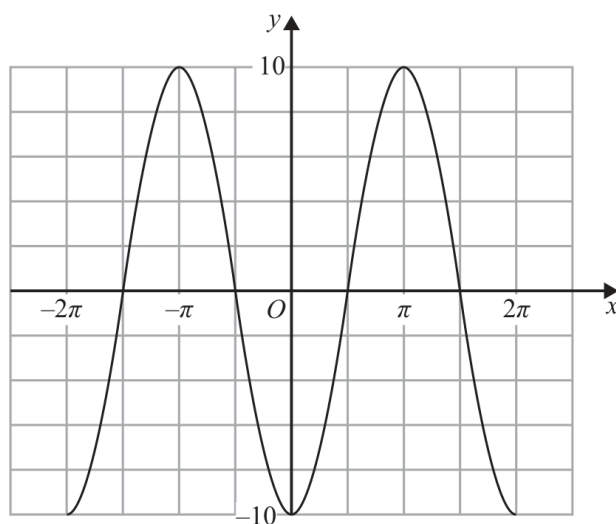


Figure 4

Figure 4 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$.

(a) Write down an expression for $f(x)$.

(2)

Copies of Figure 4 (labelled Diagram 1 and Diagram 2) can be found on the following pages.

(b) (i) On Diagram 1 sketch a graph of the curve with equation

$$y = 10 - x^2$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = 10 - x^2 \quad \text{in the interval} \quad -100\pi \leq x \leq 100\pi$$

(3)

(c) (i) On Diagram 2 sketch a graph of the curve with equation

$$y = \tan x \quad -2\pi \leq x \leq 2\pi$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = \tan x \quad \text{in the interval} \quad -100\pi \leq x \leq 100\pi$$

giving a reason for your answer.

(4)

OBJECT A

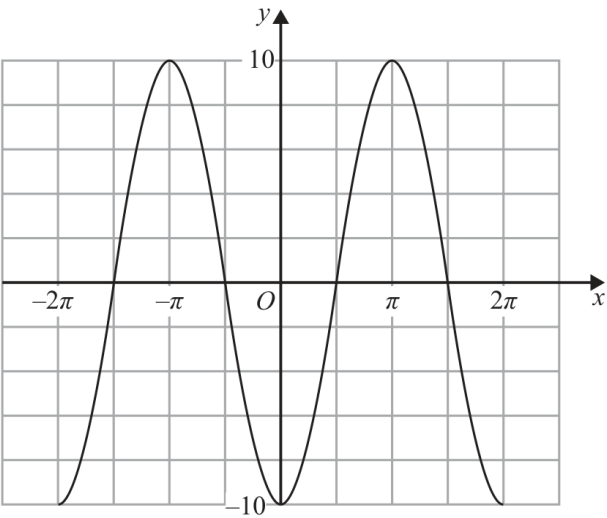


Diagram 1

OBJECT B

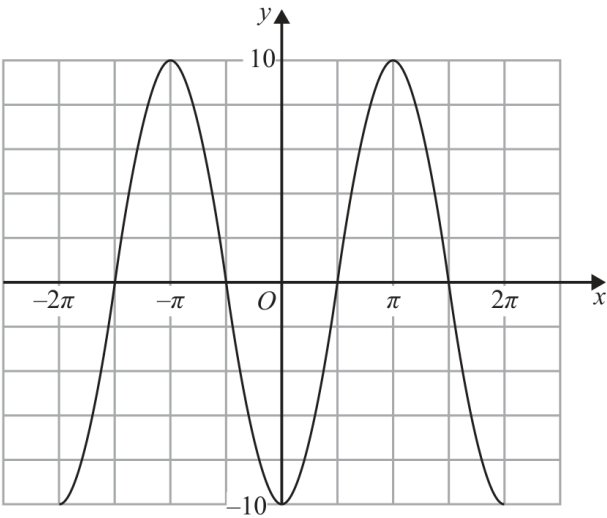


Diagram 2

(Total for Question 9 is 9 marks)

Question 15

WMA11/01 January 2026

T11 Trigonometric Functions

F03 Counting roots and graphical intersections

V02 Count intersections with a line or non-trigonometric curve

1

Identify $f(x)$

The graph has amplitude 10, period 2π , and value -10 at $x = 0$, so $f(x) = -10 \cos x$.

2

Sketch y equals 10 minus x squared

The graph $y = 10 - x^2$ is a downward-opening parabola with vertex $(0, 10)$ and x -intercepts at $x = \pm\sqrt{10}$.

3

Count intersections with the parabola

On the given sketch, the parabola meets $f(x)$ twice. Outside the central part the parabola is below the range of $f(x)$, so there are 2 solutions in $-100\pi \leq x \leq 100\pi$.

4

Sketch y equals $\tan x$

The graph $y = \tan x$ has zeros at integer multiples of π and vertical asymptotes at $x = \frac{\pi}{2} + n\pi$.

5

Count in the reference interval

In $-2\pi \leq x \leq 2\pi$, the graph $y = \tan x$ intersects $f(x)$ four times.

6

Scale to the required interval

The interval $-100\pi \leq x \leq 100\pi$ is 50 times as long as $-2\pi \leq x \leq 2\pi$, so the number of solutions is $50 \times 4 = 200$.

FINAL ANSWER

(a) $f(x) = -10 \cos x$. (b)(ii) 2. (c)(ii) 200.

DR ESLAM AHMED

4.

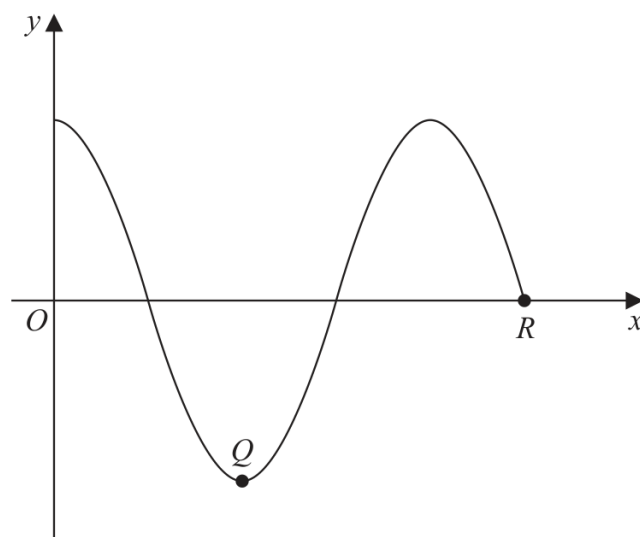


Figure 2

Figure 2 shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = \cos 2x^\circ \quad 0 \leq x \leq k$$

The point Q and the point $R(k, 0)$ lie on the curve and are shown in Figure 2.

(a) State

- (i) the coordinates of Q ,
- (ii) the value of k .

(3)

(b) Given that there are exactly two solutions to the equation

$$\cos 2x^\circ = p \quad \text{in the region } 0 \leq x \leq k$$

find the range of possible values for p .

(2)

(Total 5 marks)

Question 16

WMA11/01 October 2021

T11 Trigonometric Functions F03 Counting roots and graphical intersections and V03 Parameter values giving an exact root count

1

Find Q

For $f(x) = \cos 2x^\circ$, the first minimum occurs when $2x = 180^\circ$, so $x = 90^\circ$ and $y = -1$. Therefore $Q = (90, -1)$.

2

Find k

The point $R(k, 0)$ is the x-intercept after the next maximum. This occurs when $2x = 450^\circ$, so $x = 225^\circ$. Hence $k = 225$.

3

Count horizontal intersections

A horizontal line $y = p$ gives exactly two intersections on $0 \leq x \leq 225$ when it cuts the two negative branches only, so $-1 < p < 0$.

4

Include the maximum case

There are also exactly two solutions when $p = 1$, at $x = 0$ and $x = 180$. Therefore $-1 < p < 0$ or $p = 1$.

FINAL ANSWER

(a)(i) $Q = (90, -1)$. (a)(ii) $k = 225$. (b) $-1 < p < 0$ or $p = 1$.